KAIZEN MATHS

Lesson starter

Line and Curve Intersections

Parabola-Line Intersections

Name _____

S1

Find the intersection point:

Parabola: $y = 3x^2 - 3x - 2$

Line: $x = -2$

S2

Find the intersection points:

Parabola: $y = 3x^2 - 3x - 3$

Line: $y = 0x - 3$

S3

Find the intersection points:

Parabola: $y = 4x^2 + 4x + 3$

Line: $y = -3x + 5$

S4

Find the intersection point:

Parabola: $y = 3x^2 - 2x + 1$

Line: $x = -3$

Show your working

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Find the intersection point:

Parabola: $y = 3x^2 - 2x + 1$

Line: $x = -3$

ALG-5F81-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Line and Curve Intersections

Parabola-Line Intersections

Name _____

EA1

Find the intersection points:

Parabola: $y = 2x^2 - 4x - 1$

Line: $y = 0x$

EA2

Find the intersection points:

Parabola: $y = 4x^2 + 3x$

Line: $y = 3x + 3$

KAIZEN MATHS

Exit ticket

Line and Curve Intersections

Parabola-Line Intersections

Name _____

EB1

Find the intersection point:

Parabola: $y = 4x^2 + 3x - 3$

Line: $x = 3$

KAIZEN MATHS

Exit ticket

Line and Curve Intersections

Parabola-Line Intersections

Name _____

EA1

Find the intersection points:

Parabola: $y = 2x^2 - 4x - 1$

Line: $y = 0x$

EA2

Find the intersection points:

Parabola: $y = 4x^2 + 3x$

Line: $y = 3x + 3$

ALG-5F81-L2 · EA Show your working

KAIZEN MATHS

Exit ticket

Line and Curve Intersections

Parabola-Line Intersections

Name _____

EB1

Find the intersection point:

Parabola: $y = 4x^2 + 3x - 3$

Line: $x = 3$

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Teacher answers & working

Line and Curve Intersections · Parabola-Line Intersections

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S1

Find the intersection point:

Parabola: $y = 3x^2 - 3x - 2$

Line: $x = -2$

Answer:

One intersection point: (-2.000, 16.000)

Working:

Substitute x = -2 into the parabola

$y = 3(-2)^2 + 6 - 2$

$y = 16$

Intersection point: (-2.000, 16.000)

S2

Find the intersection points:

Parabola: $y = 3x^2 - 3x - 3$

Line: $y = 0x - 3$

Answer:

Two intersection points: (1.000, -3.000) and (0.000, -3.000)

Working:

Given equations

Parabola: $y = 3x^2 - 3x - 3$

Line: $y = 0x - 3$

Substitute the line into the conic

This gives the quadratic:

$3x^2 - 3x = 0$

Use the discriminant

$D = b^2 - 4ac = 9.000$

Solve the quadratic to find x-values: 1.000, 0.000

Substitute each x-value into y = 0x - 3 to find y.

Point 1: x = 1.000, y = -3.000, so (1.000, -3.000)

Point 2: x = 0.000, y = -3.000, so (0.000, -3.000)

Teacher answers & working

Line and Curve Intersections · Parabola-Line Intersections

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S3

Find the intersection points:

Parabola: $y = 4x^2 + 4x + 3$

Line: $y = -3x + 5$

Answer:

Two intersection points: (0.250, 4.250) and (-2.000, 11.000)

Working:

Given equations

Parabola: $y = 4x^2 + 4x + 3$

Line: $y = -3x + 5$

Substitute the line into the conic

This gives the quadratic:

$4x^2 + 7x - 2 = 0$

Use the discriminant

$D = b^2 - 4ac = 81.000$

Solve the quadratic to find x-values: 0.250, -2.000

Substitute each x-value into $y = -3x + 5$ to find y.

Point 1: x = 0.250, y = 4.250, so (0.250, 4.250)

Point 2: x = -2.000, y = 11.000, so (-2.000, 11.000)

S4

Find the intersection point:

Parabola: $y = 3x^2 - 2x + 1$

Line: $x = -3$

Answer:

One intersection point: (-3.000, 34.000)

Working:

Substitute x = -3 into the parabola

$y = 3(-3)^2 + 6 + 1$

$y = 34$

Intersection point: (-3.000, 34.000)

Teacher answers & working

Line and Curve Intersections · Parabola-Line Intersections

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EA1

Find the intersection points:

Parabola: $y = 2x^2 - 4x - 1$

Line: $y = 0x$

Answer:

Two intersection points: (2.225, 0.000) and (-0.225, 0.000)

Working:

Given equations

Parabola: $y = 2x^2 - 4x - 1$

Line: $y = 0x$

Substitute the line into the conic

This gives the quadratic:

$2x^2 - 4x - 1 = 0$

Use the discriminant

$D = b^2 - 4ac = 24.000$

Solve the quadratic to find x-values: 2.225, -0.225

Substitute each x-value into $y = 0x$ to find y.

Point 1: x = 2.225, y = 0.000, so (2.225, 0.000)

Point 2: x = -0.225, y = 0.000, so (-0.225, 0.000)

Teacher answers & working

Line and Curve Intersections · Parabola-Line Intersections

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EA2

Find the intersection points:

Parabola: $y = 4x^2 + 3x$

Line: $y = 3x + 3$

Answer:

Two intersection points: (0.866, 5.598) and (-0.866, 0.402)

Working:

Given equations

Parabola: $y = 4x^2 + 3x$

Line: $y = 3x + 3$

Substitute the line into the conic

This gives the quadratic:

$4x^2 - 3 = 0$

Use the discriminant

$D = b^2 - 4ac = 48.000$

Solve the quadratic to find x-values: 0.866, -0.866

Substitute each x-value into $y = 3x + 3$ to find y.

Point 1: x = 0.866, y = 5.598, so (0.866, 5.598)

Point 2: x = -0.866, y = 0.402, so (-0.866, 0.402)

EB1

Find the intersection point:

Parabola: $y = 4x^2 + 3x - 3$

Line: $x = 3$

Answer:

One intersection point: (3.000, 42.000)

Working:

Substitute x = 3 into the parabola

$y = 4(3)^2 + 9 - 3$

$y = 42$

Intersection point: (3.000, 42.000)